Instagram User Analytics

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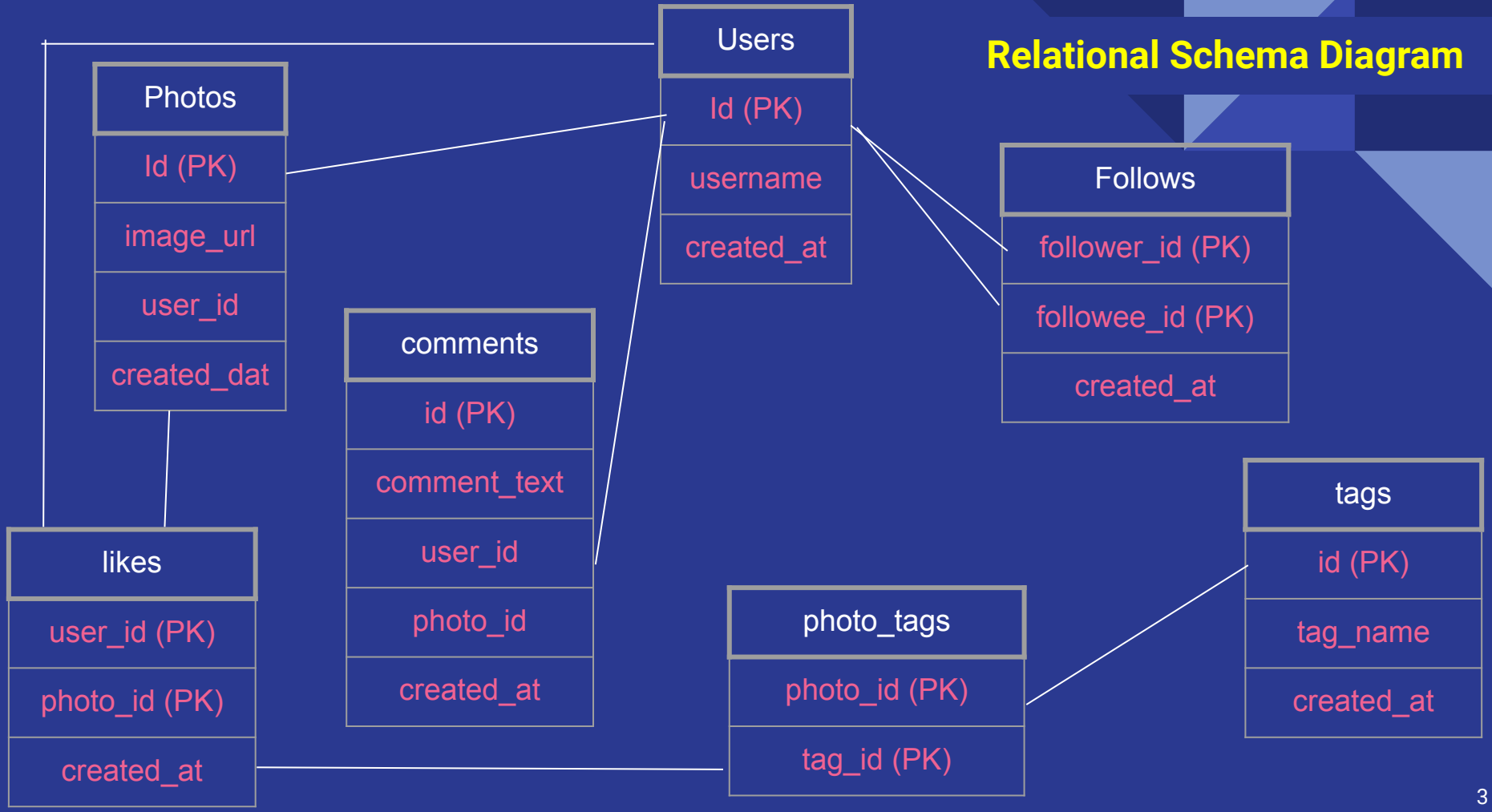
User Analysis

Marketing Analysis and Investor Metrics

Relational Schema Diagram



Marketing Analysis: 1) Loyal User Reward

Task: Identify the five oldest users on Instagram from the provided database.

Steps to get to solution -

1) Select the ‘Users’ database
2) Order the database on “created at” time column, which tells when the account was created
3) We need only 5 such users, so take top 5 such account names

Marketing Analysis: 1) Loyal User Reward

Query:

```
/* task 1 */

SELECT username, created_at
FROM users
ORDER BY created_at
LIMIT 5;
```

Output:

Result Grid | Filter Rows:

username	created_at
Darby_Herzog	2016-05-06 00:14:21
Emilio_Bernier52	2016-05-06 13:04:30
Elenor88	2016-05-08 01:30:41
Nicole71	2016-05-09 17:30:22
Jordyn.Jacobson2	2016-05-14 07:56:26

The query gives output of the five oldest users on Instagram from the given database.

Insights:

- By calculating the average number of posts per user, we can gauge overall user engagement
- A higher average indicates more active users who frequently post content
- If the average number of posts per user is low, it may suggest that users are not very active in posting content, and there might be a need for strategies to be placed and encourage more people to upload content

Marketing Analysis: 2) Inactive User Engagement

Task: Identify users who have never posted a single photo on Instagram.

Steps to get to solution:

1) Select ‘Users’ and ‘Photos’ tables, join them with id from users and User_id from photos.
2) Use left join, so all entries from Users table will get added
3) Find users in ‘Users’ table who have NULL or no entries in the corresponding ‘Photos’ table
4) Print output

Marketing Analysis: 2) Inactive User Engagement

Query and Output:

```sql
/* task 2 - join users and photos*/
SELECT u.id, u.username
FROM users u
LEFT JOIN photos p ON u.id = p.user_id
WHERE p.id IS NULL;
```

Result Grid | Filter Rows:

id	username
5	Aniya_Hackett
7	Kasandra_Homenick
14	Jaclyn81
21	Rocio33
24	Maxwell.Halvorson
25	Tierra.Trantow
34	Pearl7
36	Ollie_Ledner37
41	Mckenna17
45	David.Osinski47
49	Morgan.Kassulke
53	Linnea59
54	Duane60
57	Julien_Schmidt
66	Mike.Auer39
68	Franco_Keebler64
71	Nia_Haag
74	Hulda.Macejkovic
75	Leslie67
76	Janelle.Nikolaus81
80	Darby_Herzog
81	Esther.Zulauf61
83	Bartholome.Bernhard
89	Jessyca_West
90	Esmeralda.Mraz57
91	Bethany20

The query gives output of all the Inactive users on instagram wrt not posting any photos

Insights:

- Identifying users who have never posted can help determine the proportion of inactive or passive users
- A high number of inactive users might indicate a need for better user engagement strategies or onboarding processes to motivate users to post

Marketing Analysis: 3) Contest Winner Declaration

Task: Determine the winner of the contest and provide their details to the team.

Steps to get to solution:

1) Select ‘Users’ and ‘Photos’ and ‘Likes’ tables, join them with id from users and User_id from photos and ‘Photos’ and ‘Likes’ on photo_id column
2) Group the photo_id column and get the total number count of likes for a particular id
3) Match the photo_id and its corresponding User_id and username from ‘Users’ table
4) Arrange the columns in descending order and take the top one, so we get the one with most likes

Marketing Analysis: 3) Contest Winner Declaration

Query and Output:

```
SELECT u.id, u.username, p.id AS photo_id, MAX(photo_likes.like_count) AS max_likes
FROM users u
JOIN photos p ON u.id = p.user_id
JOIN (
    SELECT photo_id, COUNT(*) AS like_count
    FROM likes
    GROUP BY photo_id
) AS photo_likes ON p.id = photo_likes.photo_id
GROUP BY u.id, u.username, p.id
ORDER BY max_likes DESC
LIMIT 1;
```

Result Grid | Filter Rows:

id	username	photo_id	max_likes
52	Zack_Kemmer93	145	48

The query gives output of the one insta account that has gained most likes for a photo they posted

Insights:

- Identifying user who has most likes on their post, motivates them to put out good content
- It motivates others to compete for that same, not with bots/fake accounts but naturally by posting something meaningful

Marketing Analysis: 4) Hashtag Research

Task: Identify and suggest the top five most commonly used hashtags on the platform.

Steps to get to solution:

1) Select ‘Photo Tags’ and ‘Tags’ table join using tag_id column
2) Group the Tag_name in “Tag” table, and Order in descending order the count of Tag ids
3) Take top 5 entries in output

Marketing Analysis: 4) Hashtag Research

Query and Output:

```sql
/* task 4 - join photo s, photo_tags and tags,
check for count of tags used*/
select * from tags;

SELECT t.tag_name, COUNT(pt.tag_id) AS tag_count
FROM photo_tags pt
JOIN tags t ON pt.tag_id = t.id
GROUP BY t.tag_name
ORDER BY tag_count DESC
LIMIT 5;
```

Result Grid | Filter Rows:

tag_name	tag_count
smile	59
beach	42
party	39
fun	38
concert	24

The query gives output of the five most used hashtags on instagram, based on given data

Insights:

- Knowing the most commonly used hashtags can help to understand trends and user interests on the platform
- This can be valuable for brands and influencers to increase their reach
- Brands can leverage these popular hashtags in their posts to improve visibility and engagement
- It also helps in identifying trending topics.

Marketing Analysis: 5) Ad Campaign Launch

Task: Determine the day of the week when most users register on Instagram.
Provide insights on when to schedule an ad campaign.
Steps to get to solution:

1) Dayname() function to infer days from ‘created_at’ date column
2) Calculate the total number of registrations
3) Group the results on Dayname in step 1
4) Order the registrations on descending order
5) Limit the result to 1 day

Marketing Analysis: 5) Ad Campaign Launch

Query and Output:

```sql
/* task 5 -  */

SELECT
    DAYNAME(created_at) AS registration_day,
    COUNT(*) AS no_of_registrations
FROM
    users
GROUP BY
    registration_day
ORDER BY
    no_of_registrations DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Exp

registration_day	number_of_registrations
Thursday	16

The query gives output of one day where number of registrations is the most, out of other days

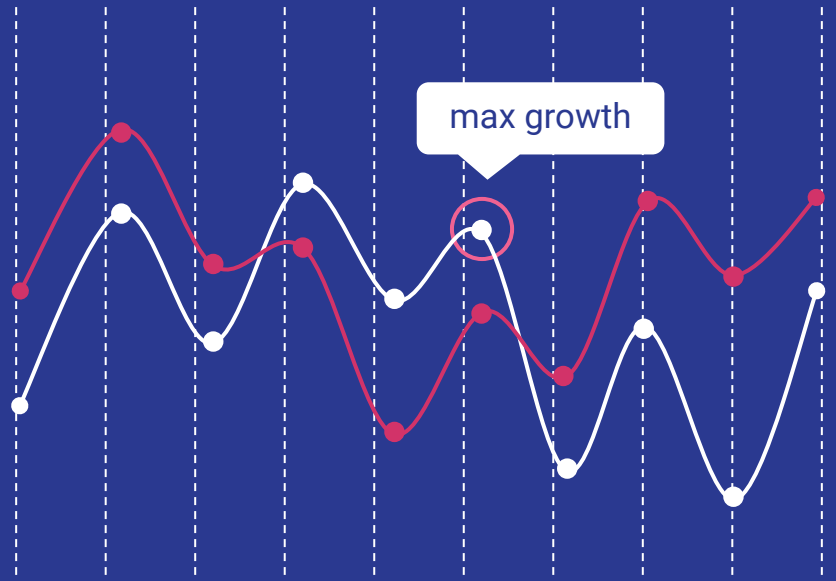
In addition:

```sql
SELECT
    DAYNAME(created_at) AS registration_day,
    COUNT(*) AS number_of_registrations
FROM
    users
GROUP BY
    registration_day
ORDER BY
    number_of_registrations DESC
LIMIT 4;
```

Result Grid | Filter Rows: | Export: | Wr

registration_day	number_of_registrations
Thursday	16
Sunday	16
Friday	15
Tuesday	14

- If we increase the limit number, we can see Sunday also has the same number of registrations as Thursday
- Also Friday is also not far behind.



Insights:

- By determining the day of the week with the highest user registrations, marketing teams can schedule ad campaigns for maximum visibility and engagement
- If user registrations peak on certain days (e.g., thursday or weekends) launching ads on those days can capture the attention of new and active users more effectively

Investor Metrics: 6) User Engagement

Your Task: Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

Steps to get to solution:

1) Count the total number of photos using ‘photos’ table
2) Count the total number of users using ‘users’ table
3) Divide the total number of photos from step (1) by the total number of users from step (2) and get the average number of posts per user

Investor Metrics: 6) User Engagement

Query and Output:

```sql
/* task 6 -  avg() and count() */

SELECT
    p.total_photos / u.total_users AS average_posts_per_user
FROM
    (SELECT COUNT(*) AS total_photos FROM photos) p,
    (SELECT COUNT(*) AS total_users FROM users) u;
```

Result Grid | Filter Row

	average_posts_per_user
▶	2.5700

The query gives output of the average number of posts that are poster by an active user

Insights:

- Platforms can assess how engaging their content is by looking at the average posts per user and inactive users.
- Implement features that encourage posting, such as challenges, contests, or improved user interface

Investor Metrics: 7) Bots & Fake Accounts

Your Task: Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

Steps to get to solution:

1) Count the total number of photos from ‘likes’ table save it using alias ‘likes_count’
2) Find users who have liked the same number of photos as the total number of photos in the ‘photos’ table
3) Group by using ‘user_id’ column in ‘likes’ table

Investor Metrics: 7) Bots & Fake Accounts

Query and Output:

```
/* task 7 -  */


SELECT l.user_id, COUNT(l.photo_id) AS likes_count
FROM likes l
GROUP BY l.user_id
HAVING COUNT(l.photo_id) = (SELECT COUNT(*) FROM photos);
```

Result Grid | Filter Rows:

user_id	likes_count
5	257
14	257
21	257
24	257
36	257
41	257
54	257
57	257
66	257
71	257
75	257
76	257
91	257

The query gives output of few accounts which might be bots or fake accounts as they have liked all posts on insta.

Insights:

- Identifying users who have liked every single photo can highlight potential bot activity
- Such behavior is not typical of normal genuine users
- Platforms can implement measures to detect and handle bot accounts, improving the quality and authenticity of user interactions
- This also helps in maintaining the platform's integrity and trustworthiness.

Thank You

Happy Instagramming!